

A
Mini-Project Report
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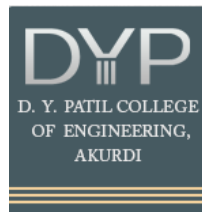
***GAS LEAKAGE PROTECTION
DEVICE AND ALERT SYSTEM***

SUBMITTED BY

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PROJECT GUIDE

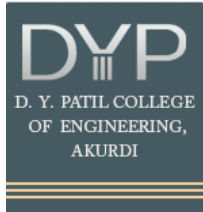
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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

**D.Y.PATIL COLLEGE OF ENGINEERING
AKURDI, PUNE – 411044**

2025-2026



D.Y.PATIL COLLEGE OF ENGINEERING
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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

CERTIFICATE

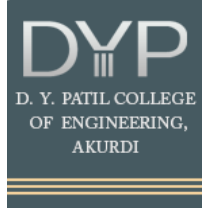
This is to certify that **Saurabh Narayan Kudale – T400080456** of T.E.
E&TC has completed the mini-project on

**‘GAS LEAKAGE PROTECTION DEVICE AND ALERT
SYSTEM’**

satisfactorily under my supervision and guidance and submitted the
project report in complete fulfillment of requirement for the award of TE
(E&TC) Degree of Engineering course under the Savitribai Phule Pune
University, Pune during the academic year 2024-2025.

Mrs.Jaya Wakode
Project Guide

Dr. Rutuja Deshmukh
H.O.D. E&TC



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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

CERTIFICATE

This is to certify that following students of T.E. Electronics & Telecommunication have completed the mini-project on '**GAS LEAKAGE PROTECTION DEVICE AND ALERT SYSTEM**' satisfactorily under my supervision and guidance and submitted the project report in complete fulfillment of requirement for the award of Bachelors Degree of Engineering course under the Savitribai Phule Pune University, Pune during the academic year 2025-2026.

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(Minimum 10 references are required as per HOD instruction)

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1. Author name, “name of paper”, name of journal, Issue no, Volume, page no.,year of publication

- Books

1. Author name, “name of book”, name of publisher, edition

INTRODUCTION

In today's modern world, the use of gases such as Liquefied Petroleum Gas (LPG), methane, and carbon monoxide has become very common in households, industries, and commercial areas. These gases are widely used for cooking, heating, and industrial processes. However, gas leakage has become a serious safety concern as it can lead to dangerous accidents such as fire, explosions, and health hazards.

Gas leakage is often difficult to detect because many gases are colorless and odorless. For example, carbon monoxide is known as a "silent killer" because it can cause severe health issues or even death when inhaled in large quantities. Therefore, it is essential to develop an efficient system that can detect gas leakage at an early stage and alert users immediately to prevent accidents.

The Gas Leakage Protection and Alert System is designed to detect the presence of harmful gases in the environment and provide an early warning to users. This system typically uses gas sensors such as MQ-2 or MQ-6 to sense the concentration of gas in the air. When the gas concentration exceeds a predefined threshold level, the sensor sends a signal to a microcontroller.

The microcontroller (such as Arduino or 8051) processes the signal and activates alert mechanisms such as a buzzer, LED indicator, or alarm system. In advanced systems, notifications can also be sent via SMS or mobile applications using GSM or IoT technology. This ensures quick response and enhances safety in both domestic and industrial environments.

The main objective of this system is to prevent accidents caused by gas leakage by providing real-time monitoring and immediate alerts. It is a cost-effective, reliable, and easy-to-implement solution that can be installed in kitchens, gas storage areas, laboratories, and factories.

This project not only improves safety but also promotes awareness about the importance of gas monitoring systems. With advancements in technology, such systems can be further enhanced by integrating smart features like automatic gas shut-off mechanisms and remote monitoring.

Literature Survey

Gas leakage detection has been an important area of research due to its significant impact on human safety, environmental protection, and industrial operations. Various techniques and systems have been developed over the years to detect gas leakage efficiently and provide timely alerts.

Early research mainly focused on traditional gas detection methods such as manual inspection, pressure monitoring, and odor-based detection. However, these methods were not always reliable or efficient in detecting small leaks or identifying the exact location of leakage. Later studies introduced sensor-based detection systems, which significantly improved accuracy and response time.

With the advancement of sensor technology, gas sensors such as MQ-series sensors (MQ-2, MQ-6, etc.) became widely used in leakage detection systems. These sensors are capable of detecting gases like LPG, methane, and carbon monoxide. Research shows that sensor-based systems are cost-effective and suitable for both domestic and industrial applications. In recent years, microcontroller-based systems have gained popularity. Platforms such as Arduino and embedded systems are widely used to process sensor data and trigger alert mechanisms like buzzers, alarms, and LED indicators. These systems provide real-time monitoring and are easy to implement, making them suitable for academic and small-scale projects.

Further advancements include the integration of Internet of Things (IoT) technology in gas leakage detection systems. IoT-based systems enable remote monitoring and alerting through mobile applications, cloud platforms, and wireless communication technologies. These systems enhance safety by providing instant notifications to users even when they are not physically present at the location.

Researchers have also explored advanced techniques such as acoustic sensing, infrared imaging, and wireless sensor networks for gas leak detection. Acoustic methods detect sound waves generated by gas leakage, while infrared imaging helps in identifying invisible gas leaks. Wireless sensor networks allow distributed monitoring over large areas, improving detection accuracy and coverage.

Recent studies focus on improving the efficiency, reliability, and accuracy of gas detection systems by using machine learning, data analysis, and smart monitoring techniques. Modern systems aim to reduce false alarms, increase sensitivity, and provide automatic control actions such as shutting off gas supply during leakage.

Despite these advancements, challenges still exist, such as sensor calibration issues, environmental interference, high installation costs for large-scale systems, and limited accuracy in detecting very small leaks. However, for domestic and small industrial applications, gas leakage detection and alert systems remain a practical and essential safety solution.

Furthermore, researchers have proposed **automated safety systems**, which not only detect gas leakage but also take preventive actions such as shutting off gas valves, activating ventilation systems, and sending emergency alerts. These systems aim to minimize human intervention and enhance safety.

Despite significant advancements, several challenges still exist, such as:

- Sensor accuracy and calibration issues
- False alarms due to environmental conditions
- High implementation cost for large-scale systems
- Limited detection range for certain sensors

However, ongoing research is focused on improving efficiency, reducing cost, and enhancing reliability. Modern gas leakage detection systems are becoming smarter, more responsive, and more suitable for real-world applications.

Problem Statement with Objectives

Gas leakage has become a major safety concern in domestic, commercial, and industrial environments due to the widespread use of gases such as LPG, methane, and carbon monoxide. Leakage of these gases can lead to severe consequences such as fire accidents, explosions, and health hazards. In many cases, gas leaks are difficult to detect at an early stage because some gases are colorless and odorless.

Traditional methods of gas leak detection, such as manual inspection or smell-based identification, are unreliable and inefficient. They often fail to provide real-time detection and immediate response, increasing the risk of accidents. Therefore, there is a need for an efficient, reliable, and automated system that can continuously monitor gas levels, detect leakage instantly, and alert users in real time.

The problem is to design and develop a **Gas Leakage Protection and Alert System** that can accurately detect gas leakage, provide immediate warning, and take necessary preventive actions to ensure safety and reduce risks. Real-time monitoring and quick response are essential requirements to avoid accidents and improve safety standards.

Objectives

- To design a system that can detect gas leakage using suitable gas sensors.
- To continuously monitor gas concentration levels in the environment.
- To develop an alert mechanism (buzzer, LED, alarm) when gas leakage is detected.
- To interface the gas sensor with a microcontroller (e.g., Arduino) for processing data.
- To provide real-time notification to users for quick action.
- To implement a system that can automatically respond to leakage (e.g., activating exhaust fan or shutting off gas supply).
- To design a low-cost, reliable, and efficient safety system for domestic and industrial use.

Specification of the Project

Hardware Requirements

The following components are required:

- ESP32 Microcontroller
 - MQ2 Gas Sensor
 - SG90 Servo Motor
 - Relay Module
 - Exhaust Fan.
 - Buzzer
 - Power Supply
-

Software Requirements

The system is programmed using the Arduino IDE.

The software includes:

- Sensor data acquisition using analog input
 - Threshold-based gas detection logic
 - Non-blocking timing using millis()
 - PWM control of servo using LEDC module
 - Relay and buzzer control via GPIO
 - IoT communication using Blynk library
-

Working Principle

Normal Condition:

Gas level is below threshold System remains idle All actuators are OFF

Gas Leakage Condition:

Step 1: ESP32 detects high gas concentration

Step 2: Servo rotates to 90° to turn OFF gas

Step 3: After 1 second, exhaust fan is turned ON

Step 4: After another 1 second, buzzer is activated

Step 5: Notification is sent to the user

Advantages

- Enhances household safety
 - Fully automatic operation
 - Real-time monitoring
 - Low cost and easy implementation
 - IoT-enabled remote access
-

Limitations

- Limited sensor accuracy
 - Environmental interference
 - False alarms
 - Detection range limitation
 - Delay in detection of small leaks
 - Power dependency
 - Maintenance requirement
 - Not suitable for all gases
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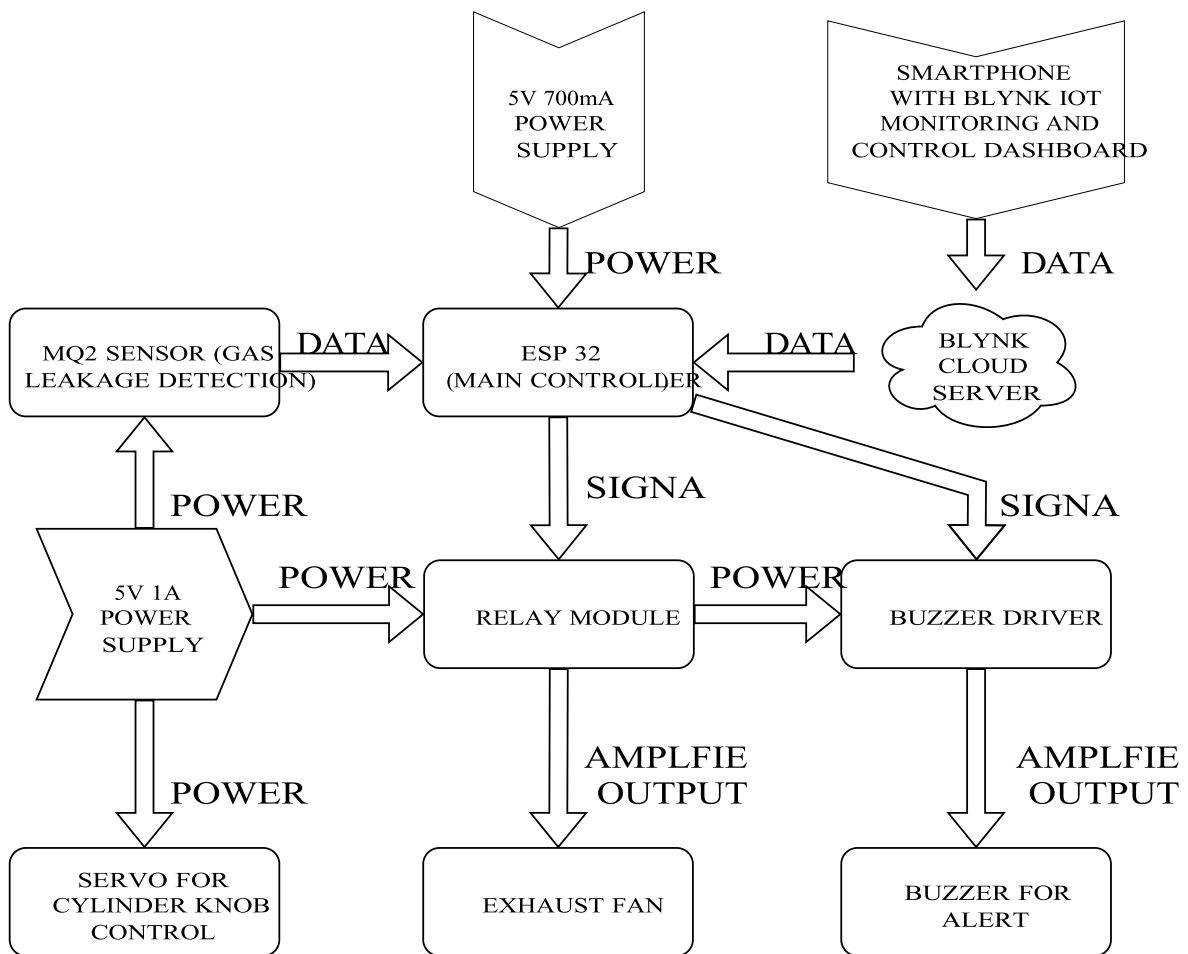
Applications

- Domestic use
 - Industrial applications
 - Commercial buildings
 - Gas storage and distribution systems
 - Laboratories
 - Smart homes and IoT systems
 - Automotive industry
 - Mining and underground environments
 - Power plants and energy sector
-

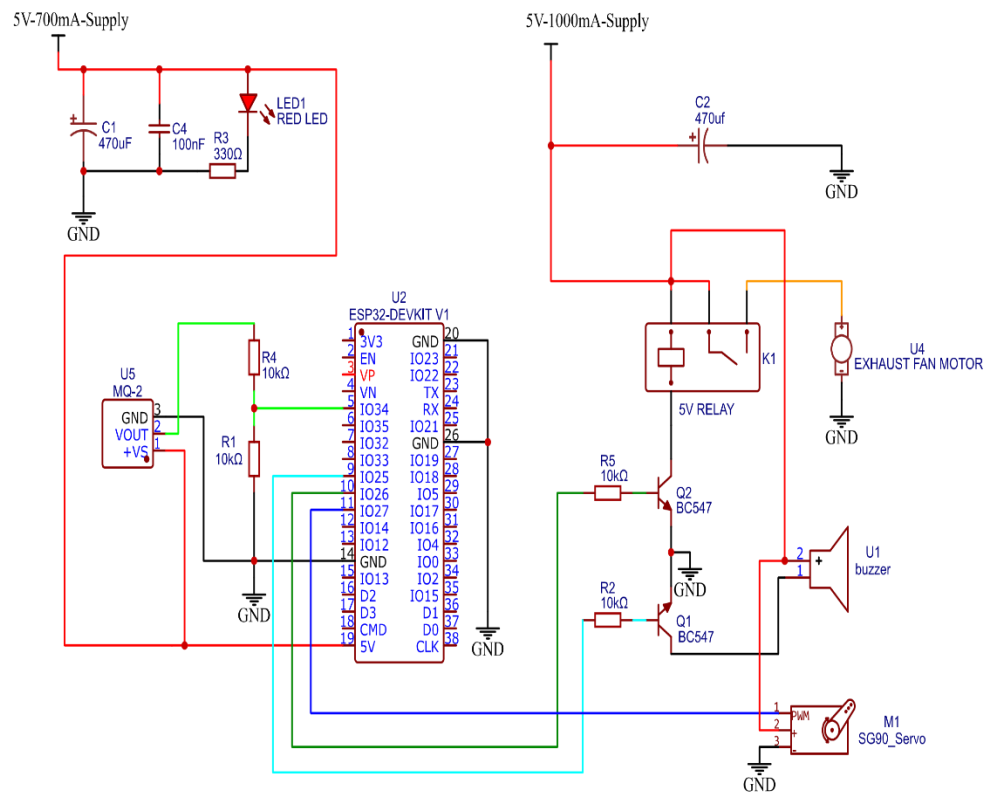
Future Scope

- Integration with IoT and Cloud Technology
- Automatic Gas Shut-off Mechanism
- Use of Artificial Intelligence (AI)
- Mobile and Robotic Detection Systems
- Smart Home Integration

BLOCK DAIGRAM:



CIRCUIT DAIGRAM:



TITLE: GAS LEAKAGE PROTECTION USING ESP32		REV: 1.0
Company: JVS TECH		Sheet: 1/1
Date: 2026-04-24	Drawn By: KIRAN AND SAURABH	

SELECTION OF COMPONENTS:

- **ESP32:**



The ESP32 is the main controller of your system, featuring a dual-core processor, built-in Wi-Fi, and multiple GPIO pins. It operates at 3.3V logic levels and supports analog and digital inputs. In your LPG system, the ESP32 reads data from the MQ-2 sensor, processes it, and decides when to trigger safety actions. It controls the buzzer, relay (fan), and servo motor. Additionally, it connects to Wi-Fi to send real-time alerts to your smartphone using platforms like Blynk. Its high performance and connectivity make it ideal for IoT-based safety systems.

- **MQ-2:**



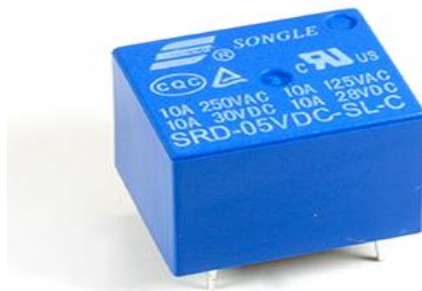
The MQ-2 is a widely used gas sensor capable of detecting LPG, propane, methane, hydrogen, and smoke. It operates at 5V and provides both analog and digital outputs. The analog output varies with gas concentration, which the ESP32 reads using its ADC pin. It requires a warm-up time (about 20–30 seconds minimum, better accuracy after a few minutes). In your system, MQ-2 continuously monitors air quality. When gas concentration exceeds a preset threshold, it signals the ESP32 to trigger safety actions like turning off the gas via servo, activating the fan, sounding the buzzer, and sending alerts.

- **Buzzer:**



A buzzer is an audio signaling device used to provide immediate audible alerts. It typically operates at 3.3V–5V and can be either active (built-in oscillator) or passive (requires PWM signal). In your LPG system, an active buzzer is preferred for simplicity. It is connected to a GPIO pin of the ESP32 and turns ON when gas leakage is detected. This provides instant local warning even if the user is not near their phone. The buzzer ensures safety by alerting occupants quickly so they can take action, especially in cases where Wi-Fi notifications might be delayed or missed.

- **5V Relay Module:**



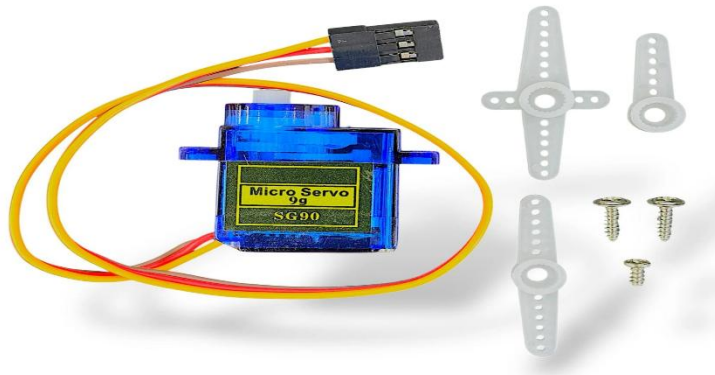
A 5V relay module is used to control high-voltage or high-current devices using low-voltage signals from the ESP32. It typically includes a relay, transistor driver, and opto-isolation for safety. The ESP32 sends a 3.3V signal to the relay input pin, which switches the internal relay to connect or disconnect the AC load. In your system, the relay controls the exhaust fan. When gas leakage is detected, the ESP32 activates the relay, turning ON the fan to expel gas. It provides electrical isolation between the low-power control circuit and the high-power fan.

- **5V Exhaust Fan**



The 5V exhaust fan is used to remove leaked gas from the environment. It operates on a DC supply and typically draws moderate current (200mA–1A depending on size). In your LPG system, the fan is activated automatically when gas leakage is detected. It helps reduce gas concentration quickly, lowering the risk of fire or explosion. The fan is controlled via a relay module, allowing the ESP32 to switch it ON or OFF safely. Proper placement of the fan near the gas source improves efficiency in removing harmful gases from the surrounding area.

- **SG90 Servo Motor:**



The SG90 is a small 5V servo motor capable of rotating approximately 0° to 180° . It is controlled using PWM signals from the ESP32. The servo provides precise position control, making it ideal for mechanical operations. In your LPG system, the servo is used to physically turn the gas regulator knob ON or OFF. When a leak is detected, the ESP32 sends a signal to rotate the servo to a predefined angle, shutting off the gas supply. It consumes up to 500–700mA under load, so a separate 5V power supply is recommended for stable operation.

- **5V 700mA Power Supply (for ESP32):**

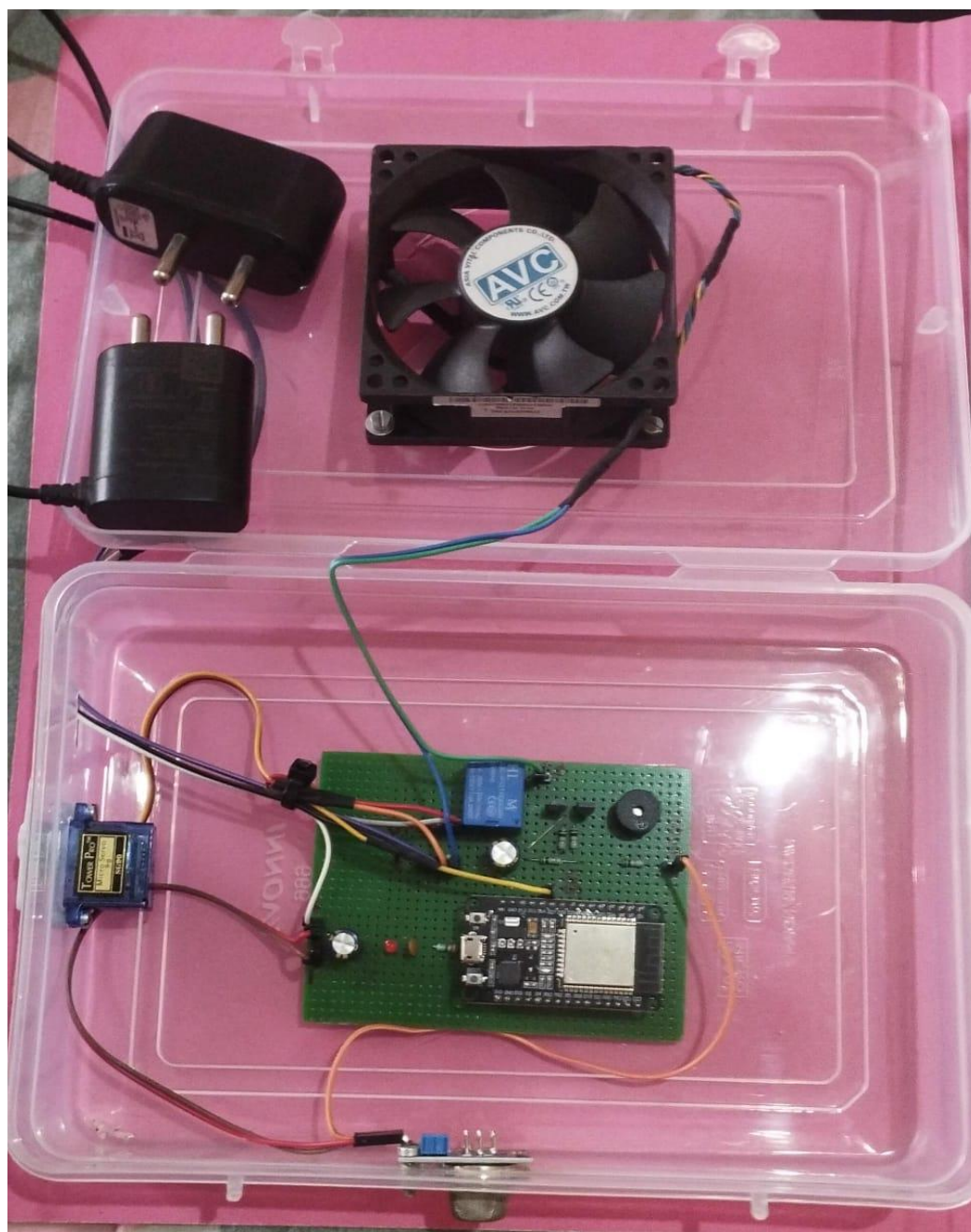


This regulated 5V 700mA supply is used to power the ESP32 safely. The ESP32 requires a stable input (via VIN or 5V pin) to operate reliably, especially when Wi-Fi is active. Insufficient current can cause random resets or unstable behavior. This supply ensures consistent voltage and enough current for normal operation. It should be well-regulated and noise-free to avoid affecting sensor readings. In your system, this dedicated supply isolates the ESP32 from high-current components like the servo and fan, improving overall stability and preventing voltage drop

- **5V 2A Power Supply (for Servo, MQ-2, Fan):**



This high-current 5V 2A supply powers components that require more current, such as the servo motor, MQ-2 sensor heater, and exhaust fan. The servo alone can draw up to 700mA under load, while the fan may require additional current. Using a separate supply prevents overload on the ESP32 power line. This ensures smooth servo movement, stable sensor operation, and reliable fan performance. It is important to connect the ground of this supply with the ESP32 ground to maintain proper signal reference and avoid erratic behavior in the system.



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